

University of Khartoum
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Assessment of Solar-Powered Irrigation Pumps in Gezira State, Sudan

A thesis submitted in partial fulfillment of the requirements for the
degree of Master of Science in Agricultural Engineering

By:

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Declaration

I, Amna Osman Elhassan, declare that this thesis is my own work and has not been submitted for any other degree.

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Dedication

To my mother Fatima Ahmed and my father Osman Elhassan, and to my brother Mohamed.

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Abstract

This study assessed the technical and economic performance of solar-powered irrigation pumps in twelve smallholder schemes of Gezira State. Pump discharge, operating hours and maintenance records were collected over two seasons. Solar pumps reduced seasonal energy costs by 61 percent compared with diesel pumps, while discharge fell during the dusty months of the dry season. The study recommends regular panel cleaning and shared maintenance arrangements among farmer groups.

Keywords: solar irrigation, pumps, Gezira, smallholders

المستخلص

هدفت هذه الدراسة إلى تقييم الأداء الفني والاقتصادي لمضخات الري العاملة بالطاقة الشمسية في اثني عشر مشروعا صغيرا بولاية الجزيرة. جمعت بيانات التصريف وساعات التشغيل وسجلات الصيانة خلال موسمين. خفضت المضخات الشمسية تكاليف الطاقة الموسمية بنسبة 61 في المائة مقارنة بمضخات الديزل. توصي الدراسة بالتنظيف المنتظم للألواح.

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Irrigated agriculture in Gezira depends on reliable pumping. Interviews with the scheme manager, Mr. Babiker Omer, informed the study design.